Blade Runner: an evidence-led path through three stages

31 levels completed · 81 required series positions · 93 valid plans. One Python application handles the full game. It reads a map and resource goal, then returns an ordered JSON plan within the game's turn limit. Advancement also requires the first submission within 30 seconds of task creation.

Quality advantage: validate locally before live acceptance. Hard local cases, fixed-baseline comparisons and independent plan replay reduced delivery risk before submission. The chosen delivery standard treated live tasks as customer acceptance runs: a missed deadline was a delivery failure, even if recovered later. Local cases were the main development laboratory. Any live gap was investigated locally, and every observed input became a regression. The aim was to handle each live input with the care expected for customer work.

Quality before submission. Evidence from real use.

Develop and challenge locally; use timed server tasks as acceptance runs.

1 Design local cases
Hard, random, impossible and forced-search controls

→

2 Compare candidates
Fixed inputs + baseline
Coverage before speed

→

3 Validate each plan
Independent raw replay
No valid plan → no submission

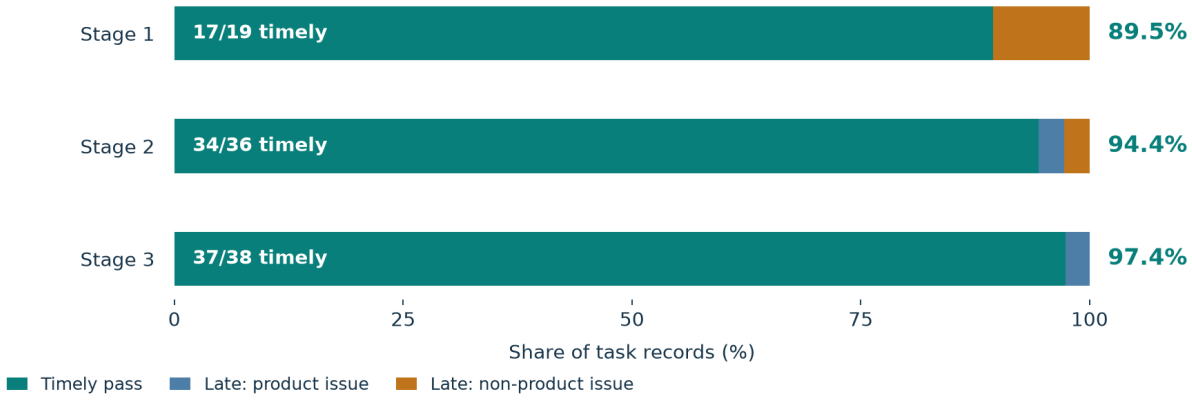
→

4 Verify real results
Submit once; check credit
Retain every input

100%
93 / 93 first submitted plans valid
No invalid answer reached the server

94.6%
88 / 93 first submissions on time
Within the 30-second task clock

2
Product issues (solver gaps)
Both stopped before submission



Five late completions: 2 product issues (solver gaps) + 3 non-product issues (task acquisition/recovery).

Stage 3 also had slow acquisition. Categories distinguish solver quality from operations, not server responsibility.

Valid first submission does not mean an immediate solve: stopped attempts were corrected locally before late completion.

Read the rates separately: all 93 plans were valid on their first and only submission; 88 arrived on time. **Product issues** mean solver gaps: two tasks (Stages 2 and 3) stopped before submission. **Non-product issues** mean task acquisition or recovery incidents: three other tasks were late. These labels distinguish solver quality from submission operations; they do not assign blame to the server. All five tasks were eventually completed; series resets lost seven earlier credits.

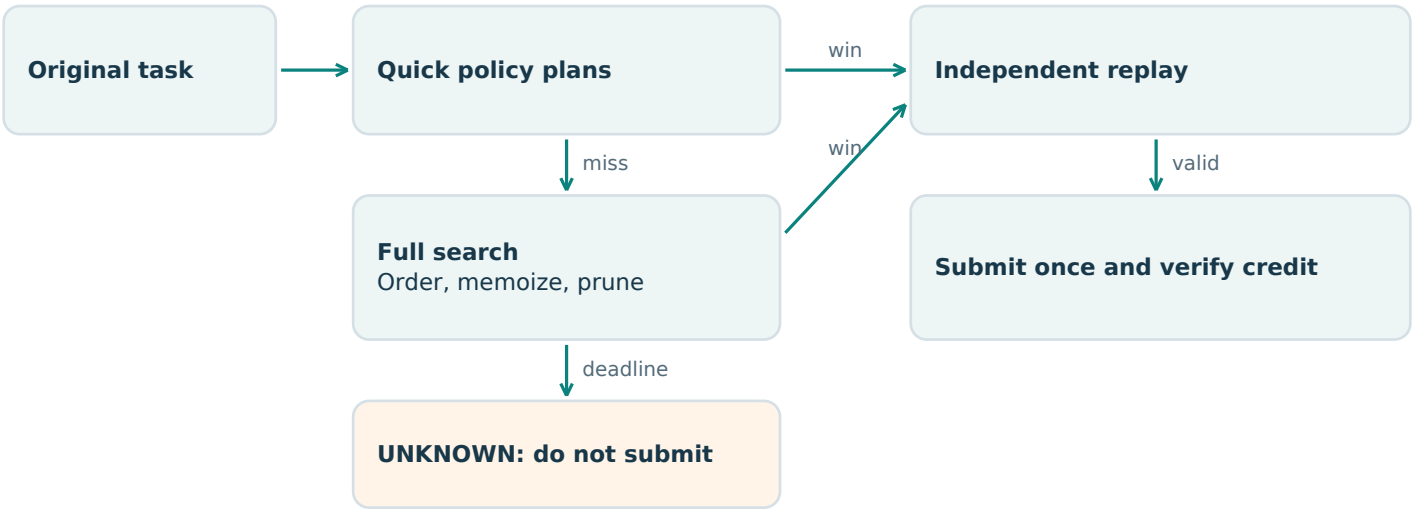
How the solver works and improves

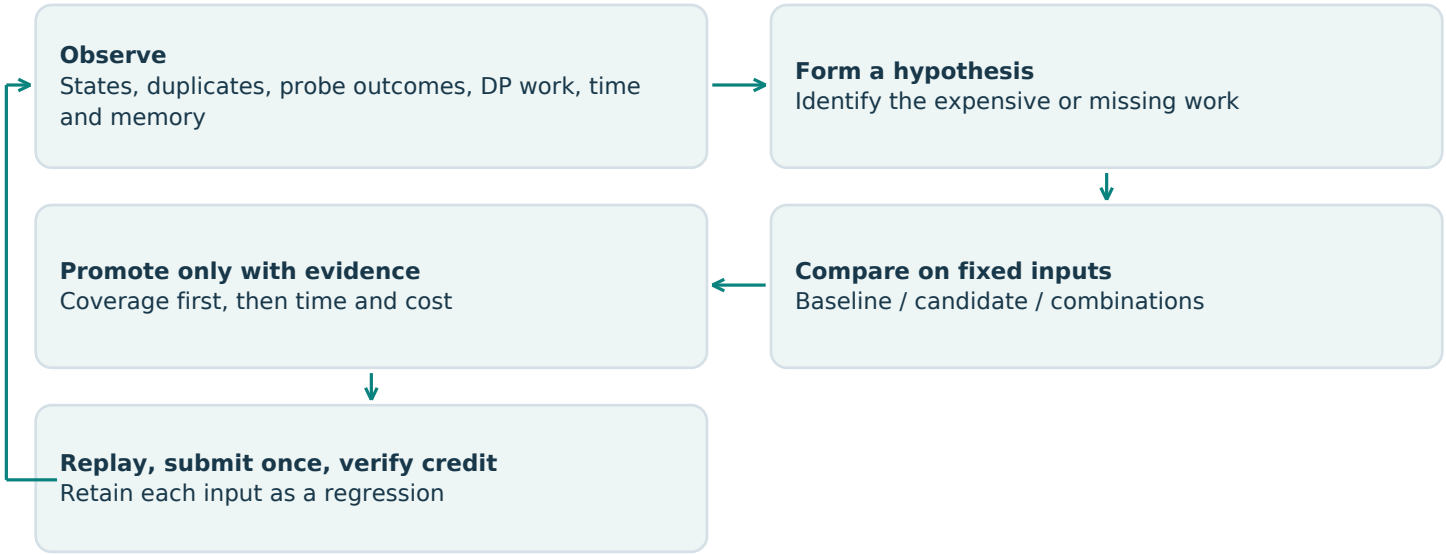
Stage 1: stone + roads
Exact search, cached geometry, investment DP

Stage 2: add ore
Reuse search; keep resources separate; saving and relaxed policies

Stage 3: add iron manufacturing
Reuse typed engine; order whole projects; optional production + on-demand DP

The shared engine tries quick policy plans, then searches remaining choices and remembers visited states. It prunes only proven dead branches. A failed quick plan leaves full search available. **DP** means saving answers to repeated funding or production subproblems.





Observability was part of the design. Search counters, timings, probe and pruning outcomes, failure reasons and saved cases made performance and errors inspectable. Building and checking that evidence took engineering time; it supported diagnosis and baseline comparisons before selecting improvements.

Stage 1 - build a correct search, then remove repeated work

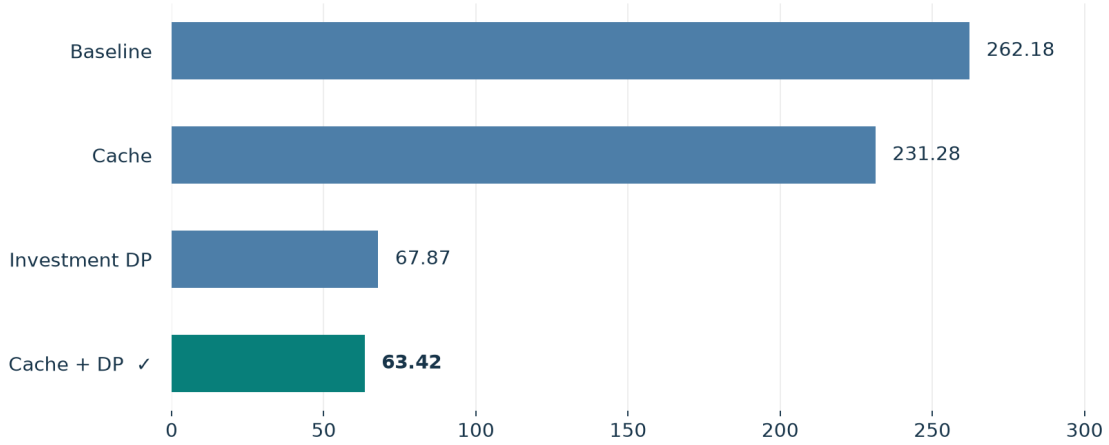
New problem. Stone is both construction money and the goal. Roads enable transport; quarries create income. A profitable final balance is insufficient: every construction action must be affordable when it executes.

Design. Model topology, inventory, turn and used activations. Search legal continuations, remember equivalent states, and try cheap completion policies. A producer and its connecting roads are one investment. Independent raw-action replay checks the resulting plan against the original map.

Observation → decision. Counters exposed repeated connectivity work. Caching reuses routes when the network is unchanged. Investment DP rejects states that cannot afford the goal even in an easier game with free roads. DP showed little benefit on easy inputs; bad-investment controls revealed where it mattered.

Stage 1 | Reuse geometry. Reject bad investments.

Decision: retain cache + investment DP after equal-outcome comparison.



75.8% less time

All four variants: 66 wins + 42 impossibility proofs

31 aligned warm rounds • 106 common cases • median of per-round solve-time sums • startup excluded

Two slow controls excluded equally from warm comparisons; all 108 retained in cold validation. No UNKNOWNs.

Adopted: cache + investment DP: **75.8% lower warm suite time**, equal outcomes. The cold corpus retains two slow cases omitted equally from warm timing.

Stage 1 delivery summary

Measure	Result
Progression	17/17 levels; 17 required series positions completed
First submitted plan valid / timely	**19/19 (100%)** / **17/19 (89.5%)**
Observed maps and starting resources	Largest 25×24; 1-23 stone sites; 12-19 starting stone
Goals and limits	15-300 stone; 4-14 turns
Timely solver time: median / maximum	0.648 / 14.181 ms
Timely server clock: median / maximum	1.481 / 2.815 s
Final local controls	108: 66 winning plans, 42 impossibility proofs, zero UNKNOWN
Late-task issues	**0 product; 2 non-product**

Integration lesson: task-ID handling and timeout recovery caused two late tasks. The corrected runner locates an existing task before creating another and verifies progression. The upstream timeout cause remains unconfirmed.

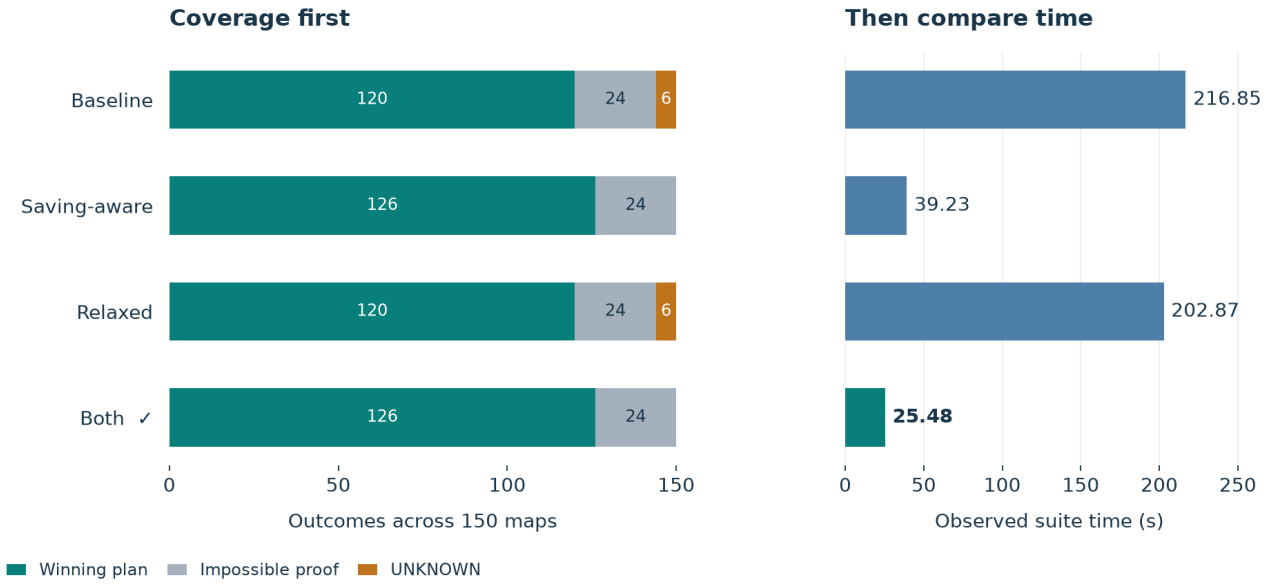
Stage 2 - preserve capital while introducing a second resource

New problem. Ore cannot pay construction costs. Building the nearest producer can spend the stone needed for a better investment. The Stage 1 search is reused with separate stone and ore inventories and resource-aware routes.

Observation → decision. A pruning/reuse candidate improved common wins by 6.5%, yet still missed the same six known-solvable cases. Across all 111 inputs, the time reduction was only 0.2%; the bundle was not adopted. The comparison paired saving-aware policies, which can wait to fund a useful project, with relaxed-resource guidance, which uses an easier economic model to suggest quick plans. These policies guide search; their failure never proves impossibility.

Stage 2 | Close the six gaps before optimizing speed.

Decision: saving-aware guidance closes the gaps; relaxed guidance makes the combination faster.



6 UNKNOWNs → 0 Both: 25.48 s vs baseline 216.85 s

One cold observation per map per variant; total pipeline time includes actual timeouts. Same 150-map corpus.

Repeated economic challenges separately supported relaxed guidance: 5/7 selected challenges met the 2× rule.

Adopted: both policies. Saving-aware guidance closed all six gaps. Relaxed guidance added no standalone coverage in this corpus. In a separate challenge, five of seven selected economic cases met a repeated 2× speed/work criterion. That evidence justified keeping it. Selected development challenges are not estimates of how often such maps occur on the server.

The product issue shows why counters matter:

Observed stop
100,000-state cap
4.247 s; no submission

→

Test the obvious fix
Remove cap only
Still UNKNOWN at 24 s

→

Inspect route decisions
Correct resource-specific ordering
Same map wins in 49.658 ms

Removing the state cap alone was insufficient. The corrected map remains a regression; the final solver uses only a time limit.

Stage 2 delivery summary

Measure	Result
Progression	7/7 levels; 30 required series positions completed
First submitted plan valid / timely	**36/36 (100%)** / **34/36 (94.4%)**
Observed maps and starting resources	Up to 25×25; 3-20 stone / 0-7 ore sites; 12-23 starting stone
Goals and limits	32 ore tasks (5-250); 4 stone tasks (25-180); 6-18 turns
Timely solver time: median / maximum	20.249 / 203.250 ms
Timely server clock: median / maximum	1.499 / 3.201 s
Final local controls	150: 126 winning plans, 24 impossibility proofs, zero UNKNOWN
Late-task issues	**1 product; 1 non-product**

Two late tasks: the product issue above and a non-product acquisition timeout with no solver execution. One long successful client run spent 21.923 s acquiring the task but only 89 ms solving it. Separate timing boundaries distinguished acquisition delay from search cost.

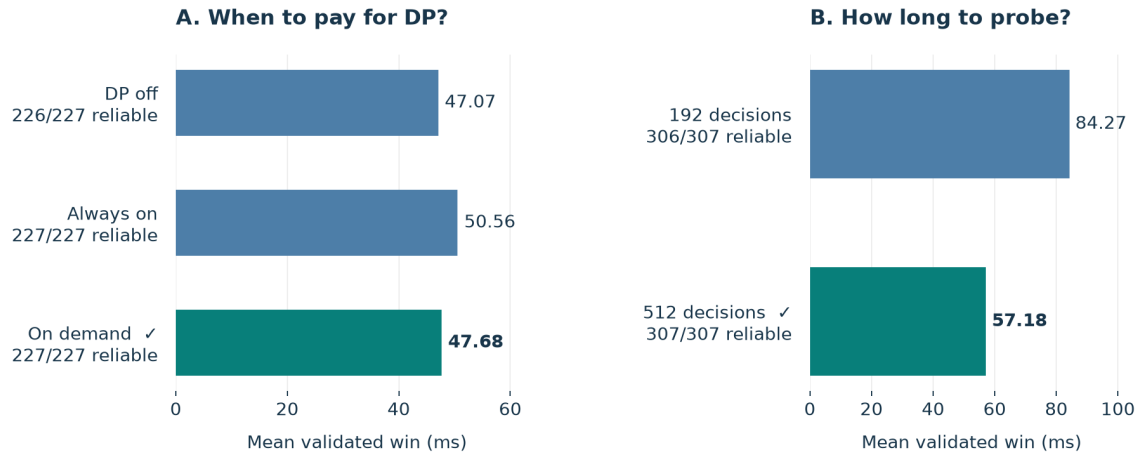
Stage 3 - schedule manufacturing and pay for stronger checks on demand

New problem. Smelters consume ore to make iron. Ore may also be required at the base, so production must stay optional. The state retains resource types and per-turn activation flags. Tasks without an iron goal can reuse earlier engines.

Design. Order whole production projects using goal gap, funds and remaining turns. Add production DP to reject impossible funding/conversion schedules, but activate it only after initial quick plans miss. The proof direction stays the same: if even the easier problem cannot win, the real branch cannot win.

Stage 3 | Pay for stronger search where it helps.

Two separate experiments: compare within each panel. Green marks the adopted option.



5.7% less time vs always on

205 common positive configurations; 227 total.
Three cold rounds. DP-off miss was an impossible case.
Initial 192 / interval 512 / later 32 held fixed.

32.1% less time vs 192

272 common positive configurations; 307 total.
Three cold rounds per arm = 1,842 runs.
Interval 512 / later 32 held fixed.

Reliable = correct in every round. Failed map: three 30 s UNKNOWNs → three wins in 0.457–0.486 s.

Stage 3 decisions and delivery

Decision A: on-demand DP. It retained all 227 reliable configurations and reduced mean winning time 5.7% versus always on. Instrumentation explains why: DP queries fell from 183,045 to 38,235 across the study; initial-probe queries fell from 144,399 to zero. Strong pruning remained available for fallback.

Not every plausible policy helped. An earlier state-triggered probe experiment on eight controls reduced expansions 6.3% but increased CPU time 21.1%. The interval trigger stayed. Counting fewer states alone would have selected the wrong improvement.

Decision B: longer initial probes. The live miss consumed 13.173 s acquiring the task, then exhausted its 11.302 s solver allowance. A local trace found a useful policy truncated at 192 decisions; allowing 256 finished it. A comparison tested 192 against 512 with the interval fixed at 512 and later probes at 32.

Across 1,842 runs, initial 512 raised reliability from 306/307 to 307/307 and reduced common winning time 32.1%. The failed map changed from three 30-second misses to three wins in 0.457-0.486 s. Two long-horizon controls slowed by 0.354/0.590 s; these costs remain in the evidence. The adopted settings are **512 initial decisions / 512-state interval / 32 later decisions** per policy.

Stage 3 delivery summary

Measure	Result
Progression	7/7 levels; 34 required series positions completed
First submitted plan valid / timely	**38/38 (100%)** / **37/38 (97.4%)**
Observed maps and starting resources	Up to 25×25; 3-21 stone / 0-7 ore sites; 12-21 starting stone
Goals and limits	30 iron tasks (5-300); 4 ore (5-100); 4 stone (25-180); 6-27 turns
Timely solver time: median / maximum	127.156 / 4,232.246 ms
Timely server clock: median / maximum	1.799 / 8.956 s
Final local controls	307 configurations: 273 winning plans, 34 impossibility proofs, zero UNKNOWN
Late-task issues	**1 product; 0 non-product**; acquisition delay also contributed to the product case

After correcting the solver gap, all 18 fresh tasks counted, at 1.218-3.537 s server time. The old task was completed late before the series restarted. Slow acquisition contributed to that incident; the original solver also missed a separate full 30-second local run.

Correctness, validation and reviewability

What the delivered code demonstrates

Client concern	Reviewable evidence
Correctness and requirements	Ordered-action replay; exact state identity; optional manufacturing; proven optimistic cuts. Full search is complete if exhausted under the declared model; deadline exhaustion returns UNKNOWN.
Validation before acceptance	Tiny exhaustive oracles, known witnesses, impossible/random/adversarial cases, tight resources, shared roads and forced full-search controls. All 93 live records / 92 distinct inputs retained.
Maintainable Python	One application for all stages; documented, typed functions. A public solve entry calls the stage engines; independent validators check plans; a separate runner handles requests, deadlines and submission.
Delivery verification	Delivery migration: 658/658 regressions, 115 portable tests and strict type checks passed; 367 historical tests also passed. A fresh package passed tests and all three offline smokes. The later API cleanup passed 116 tests and all 93 saved-task regressions with unchanged plans.
Independent judgment	Retain negative results and regressions; compare combined candidates; distinguish solver time from acquisition and server task age.

Why the search is correct within its model. Collecting available free mining income at the base early preserves later construction options. Manufacturing stays optional because it consumes ore. The search remembers the full map, stocks, turn and used production actions, so merging identical states preserves future choices. Every real win must also work in the easier game used by a pruning check; failure there safely rejects the real branch. Search order and failed quick probes remove no legal alternatives. Exhausting the finite search proves no solution exists; hitting the deadline returns UNKNOWN. Independent replay checks every returned action and the final goal.

These arguments assume up/down/left/right adjacency, renewable extraction of five units per turn, construction paid from the base, matching-resource relay rules and unlimited storage. Time grows with visited states, work per state and optional probes/DP; memory grows with stored states and caches. Both can grow exponentially. Local coverage establishes neither a universal 30-second guarantee nor correctness under different server rules.

Summary timing covers timely tasks only. Solver times exclude startup, network and the runner's outer replay. Stage 2/3 timers include internal replay; Stage 1's greedy path can include it too. Server time runs from task creation to the valid submission. Different inputs and timer scopes prevent cross-stage speedup claims. Live inputs had no mixed resource goals; the local controls cover additional combinations.

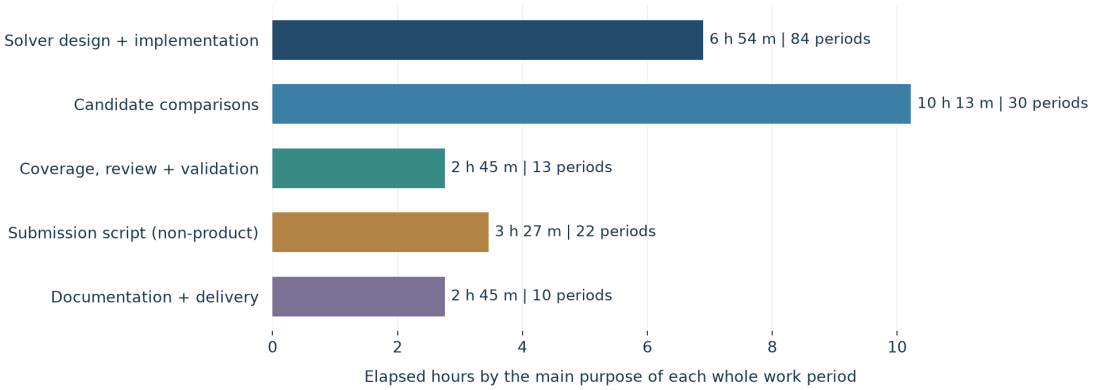
Time accounting: development, comparison and validation

Work covered **five delivered areas**: the stone search; the ore extension; manufacturing; the submission/recovery script; and the portable regression and review package. The submission script is **non-product support**, separate from the solver. Beyond implementing these, the work deliberately included candidate comparisons, adversarial controls and retained negative results. This also included building observability: instrumentation, benchmark reporting and failure diagnostics that supplied evidence for solver improvements. Passing the observed game and establishing evidence for future changes are different scopes.

Time by work area

26 h 05 m displayed assistant elapsed | 159 periods | through 18 October 2026, 20:52 (UTC+8)

Review is quality work; submission-script development and operation form one non-product category



Whole-period assignments include associated coding, checks and waits; labels are rounded.

Displayed 26:05:07.749 + excluded setup / interruption 00:12:12.908 = original ledger 26:17:20.657.

Non-product means submission support outside the solver. These are not human-labor or TiMaS hours.

The table groups **159 whole assistant work periods** by their main deliverable. Design, implementation and diagnosis belong together. Dedicated reviews join coverage and validation as quality work. Submission-script development and live operation form one non-product category.

Primary purpose	Periods	Recorded elapsed (h:mm:ss)
Solver design and implementation, including requirements and diagnosis	84	6:53:39
Comparing candidates, including experimental code and repeated runs	30	10:13:15
Quality: coverage, review and validation	13	2:45:25
Submission script: development, execution and recovery **(non-product)**	22	3:27:26
Documentation, packaging and delivery	10	2:45:23
Displayed subtotal	**159**	**26:05:08**

Comparisons and quality work account for 12 h 59 m. The quality row includes **30 m 39 s of dedicated reviews** previously grouped with development. Reviews embedded in other implementation or comparison periods cannot be timed separately. Whole periods include related coding, tests and waits; combined activities are assigned to their main deliverable, with the original classification and uncertainty retained in the audit. These are not exclusive coding or test-runtime totals. The script row includes operation and recovery, not just script coding. No time is counted twice.

Observability work is included in these existing periods under their main deliverable, not added as another time category. The ledger does not isolate its hours, so no separate total is claimed.

The displayed subtotal excludes **10:52.224** of environment/workflow handling and an interrupted period with no recorded work (**1:20.684**). Adding those to **26:05:07.749** reconciles to the unchanged **26:17:20.657** original ledger. Displayed rows are rounded independently. The original stage totals remain approximately 5 h 23 m / 6 h 58 m / 11 h 35 m, plus shared and mixed-stage work.

Scope, follow-ups and references

The game-completion turn ends after **24:15:44** recorded elapsed; later reporting and delivery add **2:01:37** through the frozen cutoff of 18 October 2026, 20:52 (UTC+8). Tools, tests, waits and agent coordination are included. Gaps and later report revisions are excluded; parallel agents are not added again. This measures assistant-turn elapsed, not human labor or TiMaS time.

What the extra scope bought. Baseline/candidate/combined comparisons exposed coverage gaps, measured speed gains and rejected misleading improvements. The Stage 3 initial-probe study alone used 1,842 runs; its adopted setting removed a known miss while recording the cases that slowed. Saved inputs, raw-action replay and reproducible checks now let future changes be evaluated before live acceptance. These are concrete quality assets; future maintenance savings have not been measured.

How to compare the time fairly. Recruitment feedback on 20 October cites about 21 tracked hours against roughly 3 hours for internal developers and 3-6 hours for candidates. That figure has not been reconciled with a TiMaS export and cannot be replaced by this differently measured ledger. The original brief described time as supplementary, but also set an expectation of a few hours. The broader comparison program was a deliberate scope choice, not an unstated requirement. The evidence supports its measured outcomes; it does not show that every experiment was necessary, that faster solutions had lower quality, or that skipping the work would have produced this solver in a few hours. Under a tighter budget, retain correctness checks and representative regressions, agree the reliability target, and timebox optional tuning.

Remaining solver work

- Confirm uncertain adjacency, mining and transfer rules.
- Compare route-cache reuse with the adopted guidance.
- **Use greedy-probe results to guide fallback search:** test whether partial progress can identify promising branches, guide the fallback policy and search order, and avoid repeating calculations. Probe results provide hints; a probe miss must never mark a state impossible or remove legal choices. Compare time to a validated win and added overhead against the current solver.

These are recorded follow-ups, with no unmeasured benefit claimed.

The delivery ZIP contains code, locked dependencies, reproducible offline commands, tests, this report and compact supporting evidence. Charts compare fixed inputs and retain failed outcomes; saved measurements and file checksums make the results traceable. Credentials and raw private conversation are excluded.

References

These sources informed the design and experiments. The game's pruning proofs and chosen settings were derived and tested for this solver; the cited planners and optimization packages are not runtime dependencies.

Source consulted	Contribution to this solver
David Churchill and Michael Buro (AIIDE, 2011), Build Order Optimization in StarCraft.	Treat construction as a schedule constrained by available resources; prioritize useful investments while preserving fallback search.
Jörg Hoffmann, Metric-FF: algorithmic principles, author documentation.	Use easier resource-planning problems to guide choices. In this solver, heuristic failure never proves impossibility or removes a legal action.
Amanda Jane Coles et al. (JAIR, 2013), A Hybrid LP-RPG Heuristic for Modelling Numeric Resource Flows in Planning.	Account for production and consumption together: spending ore on iron changes what remains for other goals. This informed the funding/production DP, without importing an LP solver.
David Tolpin et al. (IJCAI, 2013), Towards Rational Deployment of Multiple Heuristics in A*.	Defer expensive reasoning until it is useful. This motivated testing production DP after initial probes miss; the implementation retains its existing depth-first search.
SCIP Optimization Suite, How to add primal heuristics, official documentation.	Separate probe priority, frequency and timing; compare the cost of unsuccessful probes against earlier winning plans. Final intervals came from local experiments.
Chris Jones, Effective Troubleshooting, Google SRE.	Start from measured symptoms, test one explanation at a time and distinguish search cost from acquisition delay before choosing a correction.